

Experiment No: 4

AIM: To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

THEORY:

Blockchain is a **distributed and tamper-resistant digital ledger** used to record transactions and data. In this experiment, blockchain technology is used to implement a **Digital Warranty and Product Ownership Transfer System**.

The system uses a **Solidity smart contract** deployed on the **Ethereum Sepolia Testnet** through **Remix IDE**. MetaMask acts as the blockchain wallet and is used to connect the user's account with Remix and authorize transactions.

The smart contract stores product information such as:

- Product ID
- Product name
- Owner's blockchain address
- Warranty period
- Purchase date
- Product existence status

The contract also provides functions to register a product, check its warranty status, retrieve product information, and transfer ownership.

A major blockchain feature demonstrated in this project is **access control through the smart contract**. The `transferOwnership()` function contains a condition that allows only the current owner to transfer the product:

```
require(  
    products[_productId].owner == msg.sender,  
    "Only current owner can transfer"  
);
```

Therefore, when Account 1 attempted to transfer the product after ownership had already been transferred to Account 2, the transaction was rejected with:

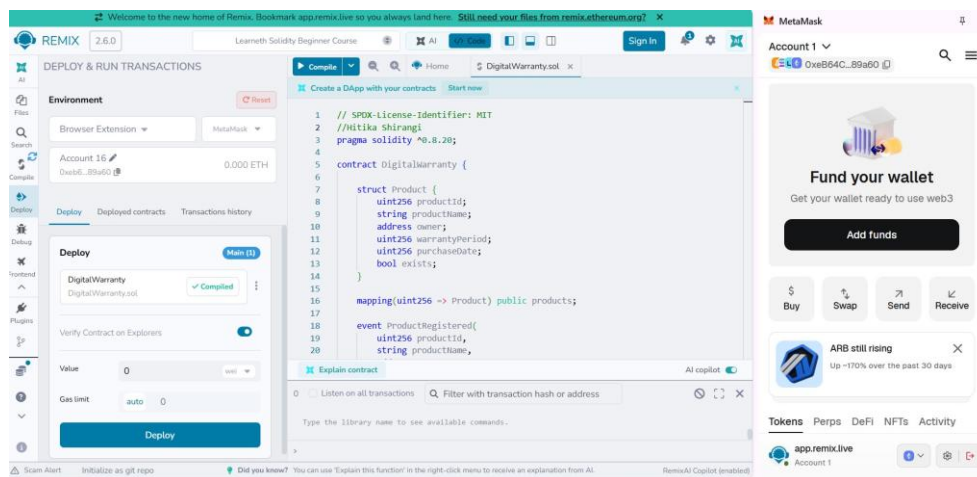
Only current owner can transfer

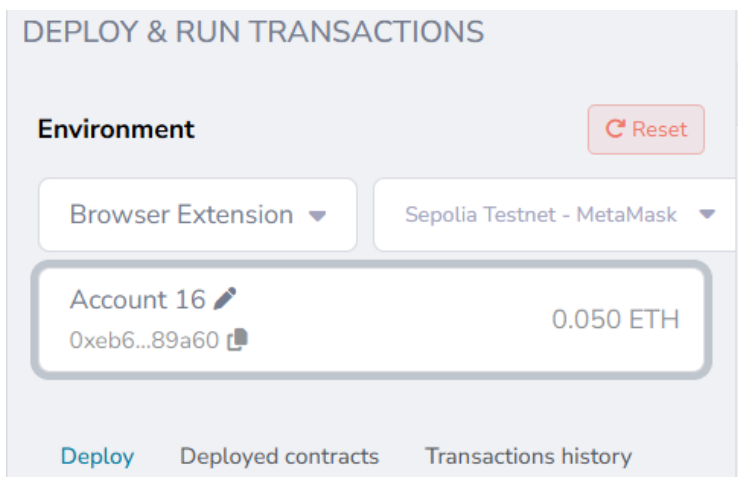
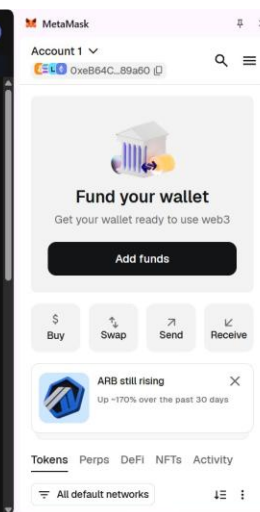
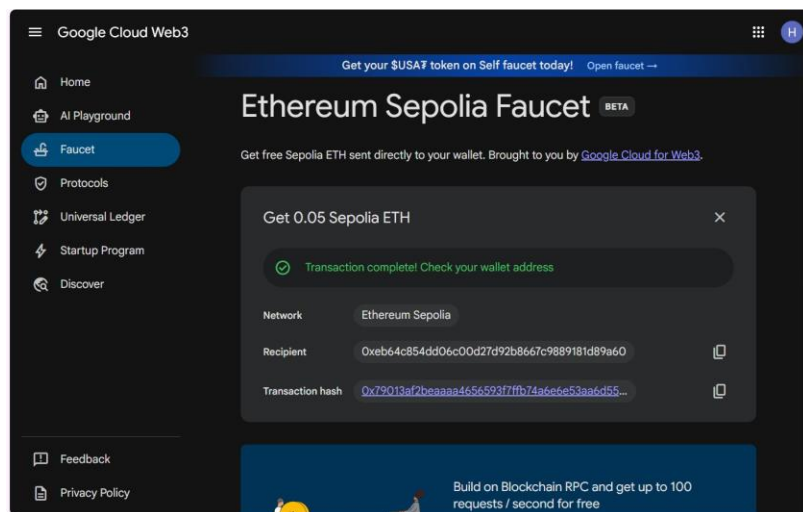
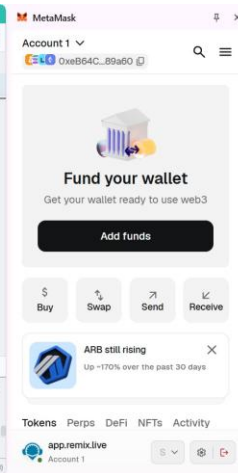
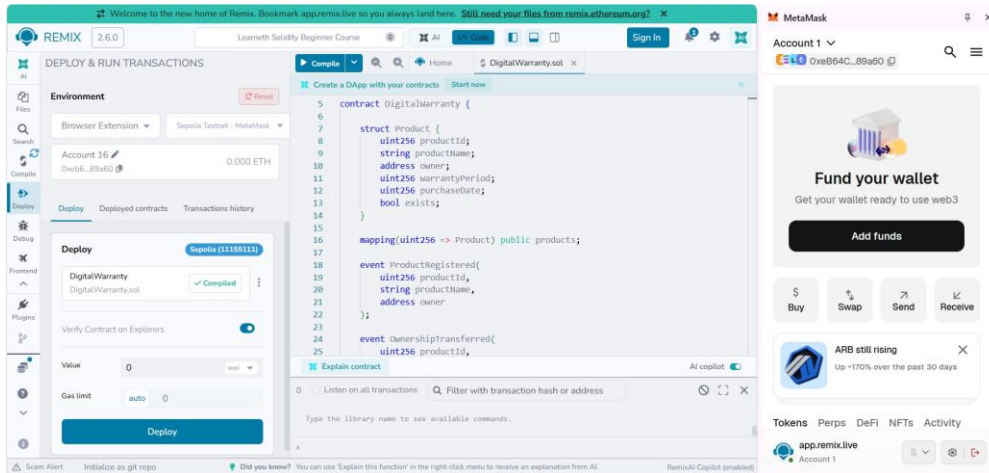
This demonstrates that the smart contract automatically enforces the ownership rule.

The experiment follows the lab requirements of setting up MetaMask, connecting a testnet to Remix, creating and deploying a Solidity smart contract, checking transaction details, and interacting with the deployed contract.

STEPS and OUTPUT:

1. Setup metamask and connect with Remix IDE
2. Create the solidity smart contract
3. Compile and deploy it.
4. Register a product
5. Verify product details
6. Check warranty
7. Transfer product ownership
8. Verify new ownership
9. Demonstrate unauthorized transfer restriction





REMIX2.6.0

Welcome to the new home of Remix. Bookmark app.remix.live so you always land here. [Still need your files from remix.ethereum.org?](#)

Learneth Solidity Beginner Course

AI

5% Tools

Sign In

DEPLOY & RUN TRANSACTIONS

Environment

Account 160xb6...89a600.050 ETH

Deploy

Deployed contracts

Transactions history

Deploy

DigitalWarranty

DigitalWarranty.sol

Verify Contract on Explorers

Value0wei

Gas limitauto0

Deploy

Compile

Create a DApp with your contracts

DigitalWarranty.sol

contract DigitalWarranty {
 struct Product {
 uint256 productId;
 string productName;
 address owner;
 uint256 warrantyPeriod;
 uint256 purchaseDate;
 bool exists;
 }
 mapping(uint256 -> Product) public products;
 event ProductRegistered(
 uint256 productId,
 string productName,
 address owner
);
 event OwnershipTransferred(
 uint256 productId,
 address owner
);
}

AI copilot

0 Listen on all transactions

Filter with transaction hash or address

Type the library name to see available commands.

Scam AlertInitialize as git repoDid you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI.

MetaMask

Account 1

Deploy a contract

This site wants you to deploy a contract

Estimated changesNo changes

NetworkSepolia

Request fromapp.remix.live

Network fee0.0025 SepoliaETH

SpeedMarket ~12 sec

CancelConfirm

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

OverviewState

TRANSACTION ACTION

Contract Created at 0xb6CC53e1...97027a52e

[This is a Sepolia Testnet transaction only]

Transaction Hash:

0xfbb445d4875fb5ae56b9713205a717537a15a962b8d59d29e800d431e0304dda

Status:

Success

Block:

117313561 Block Confirmation

Timestamp:

18 secs agoSep-18-2026 02:47:00 PM +UTC

From:

0xb64C854dD06c00D27d92B8667C9889181d89a60

To:

[0xb6cc53e1b91bb7d7185f5955e4d764297027a52e Created]

Value:

0 ETH

Transaction Fee:

0.002423563735024908 ETH

Gas Price:

2.490779922 Gwei (0.000000002490779922 ETH)

More Details:

+ Click to show more

Functions

● **checkWarranty** (uint256)

Call

_productId (uint256)

Calldata  Params 

● **getProduct** (uint256)

Call

_productId (uint256)

Calldata  Params 

● **products** (uint256)

Call

param0 (uint256)

Calldata  Params 

● **registerProduct** (uint256, string, uint256)

_productId (uint256)

_productName (string)

_warrantyPeriod (uint256)

Calldata  Params 

Transact

● **transferOwnership** (uint256, address)

_productId (uint256)

_newOwner (address)

Calldata  Params 

Transact

DEPLOY & RUN TRANSACTIONS

Environment

Browser Extension

Sepolia Testnet - MetaMask

Account 16

0xeb6...89a60

0.047 ETH

Deploy

Deployed contracts

Transactions history

products (uint256)

param0 (uint256)

Calldata Params

registerProduct (uint256, string, uint256)

1001

Samsung Refrigerator

365

Calldata Params

Transact

transferOwnership (uint256, address)

_productId (uint256)

_newOwner (address)

Compile

Create a DApp with your contracts

Start now

```
5 contract DigitalWarranty {
6
7     struct Product {
8         uint256 productId;
9         string productName;
10        address owner;
11        uint256 warrantyPeriod;
12        uint256 purchaseDate;
13        bool exists;
14    }
15
16    mapping(uint256 => Product)
17
18    event ProductRegistered(
19        uint256 productId,
20        string productName,
21        address owner
22    );
23
24    event OwnershipTransferred(
25        uint256 productId,
```

Explain contract

AI copilot

0 Listen on all transactions

AT THE START OF THE TRANSACTION, THE AI WILL TRY TO EXPLAIN IT

Account 1

Transaction request

Balance changes

No changes

Added protection

Optional

Because you're interacting with an unknown address, protection can prevent some malicious transactions.

Learn more

Network

Sepolia

Request from

app.remix.live

Interacting with

Ox6BCC5...7a52E

Network fee

0.0008 SepoliaETH

Speed

Market ~12 sec

Cancel

Confirm

getProduct (uint256)

Call

1001

Calldata Params

0:uint256: 1001

1:string: Samsung Refrigerator

2:address: 0xeB64C854dD06c00D27d92B8667C9889181d89a60

3:uint256: 365

4:uint256: 1789743204

5:bool: true

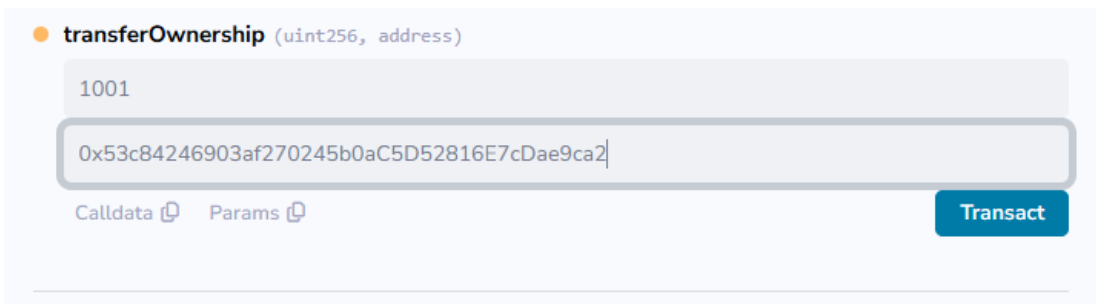
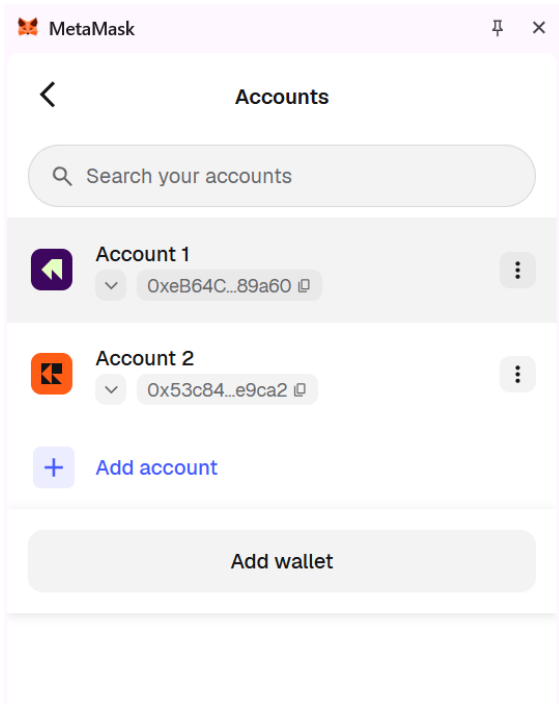
checkWarranty (uint256)

Call

1001

Calldata Params

0:bool: true





Account 1



Transaction request

Balance changes ⓘ

No changes

ⓘ Added protection ⓘ

Optional



Because you're interacting with an unknown address, protection can prevent some malicious transactions.

[Learn more](#)

Network

S Sepolia

Request from ⓘ

app.remix.live

Interacting with ⓘ

0x6BCC5...7a52E

Network fee ⓘ

0.0004 S SepoliaETH

Speed

Market ~12 sec

Cancel

Confirm

● **getProduct** (uint256)

Call

1001

Calldata ⓘ Params ⓘ

0:uint256: 1001

1:string: Samsung Refrigerator

2:address: 0x53c84246903af270245b0aC5D52816E7cDae9ca2

3:uint256: 365

4:uint256: 1789743204

5:bool: true

